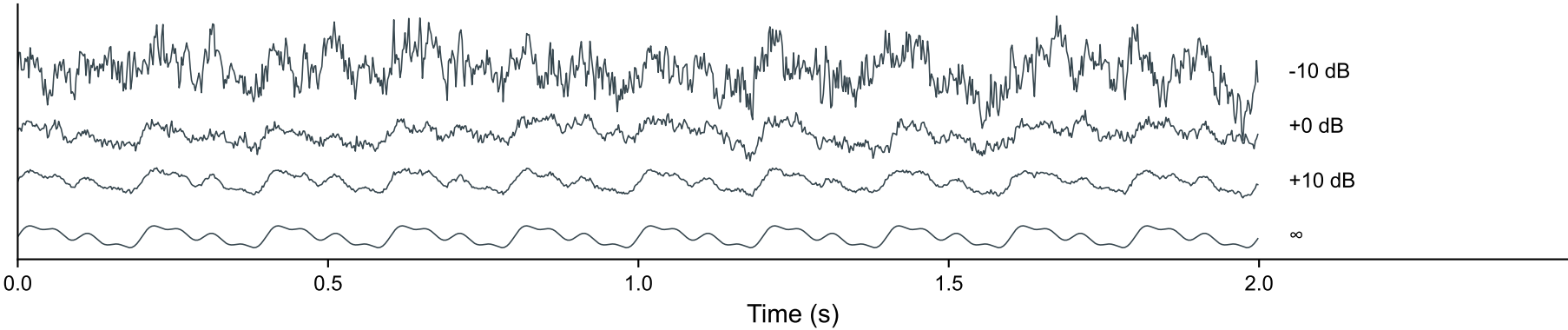
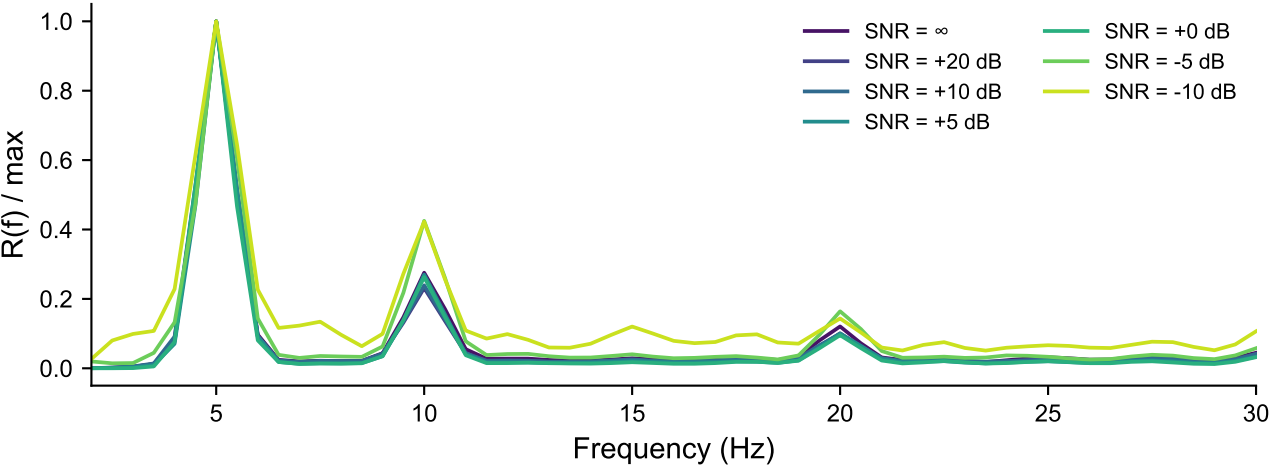


Figure 14 — SNR robustness: $R(f)$ peak structure stable down to ~ 0 dB; peak detectability falls $\sim 100\times$ from clean to -10 dB

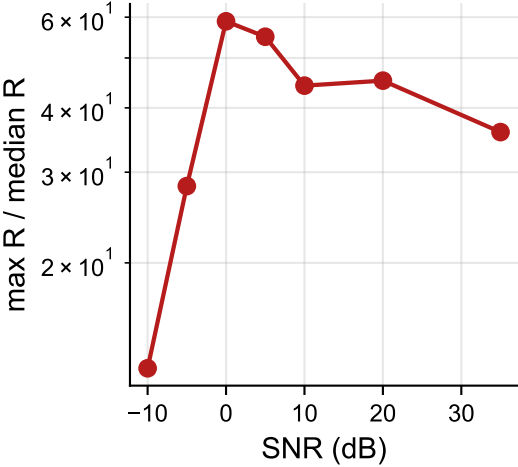
a) Same clean harmonic stack + pink noise at varying SNR (top = ∞ dB, bottom = -10 dB)



b) $R(f)$ (peak-normalized) across SNRs — peak structure preserved down to ~ 0 dB



c) Peak-detectability curve



d) Detected peak frequencies vs SNR — true carriers at 5/10/20 Hz (grey lines)

